

# Experimenting with Aesthetics-Based Graph Layout

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## 1 Introduction

Many automatic graph layout algorithms have been implemented to display relational data in a graphical (usually node-arc) manner. The success of these algorithms is typically measured by their computational efficiency and the extent to which they conform to aesthetic criteria (for example, minimising the number of crossings, maximising symmetry). Little research has been performed on the usability aspects of such algorithms: do they produce graph drawings that make the embodied information easy to use and understand? Is the computational effort expended on conforming to the assumed aesthetic criteria justifiable with respect to better usability? This paper reports on usability studies to investigate automatic graph layout algorithms with respect to human use.

## 2 Syntactic Performance

The first two experiments investigated human performance with respect to subjects' ability to answer abstract questions about the structure of a graph (e.g. How long is the shortest path between two given nodes?)

### 2.1 Individual Aesthetics

The first study considered the effects of five individual aesthetics: *minimize bends*, *minimize edge crossings*, *maximise the minimum angles between adjacent edges from a node* *orthogonality* and *symmetry*.

For each aesthetic being considered, two drawings of the same graph were produced (representing a strong or weak presence). The experiment was conducted online: subjects had the relevant terminology and the three questions explained, and a worked example was presented. They then answered three graph theoretic questions chosen as a means of measuring syntactic performance on the sixteen graph drawings.

Two dependent variables were recorded: the response time to answer each question, and the correctness of each answer. A within-subjects analysis method was used. Fifty-five second-year computer science students took part.

ANOVA analysis and Tukey's WSD pairwise comparison procedure were performed for both errors and response time.<sup>1</sup> The results indicate support for reducing the number of crossings and bends, and increasing the display of symmetry. However, no support was found for maximising the minimum angle or increasing the orthogonality.

## 2.2 Algorithms

The second study considered the effects of eight different graph drawing algorithms, all implemented in the GRAPHED system (Himsolt, 1990)). The algorithms included three force-directed algorithms, three planar orthogonal grid algorithms with bends minimisation, a planar grid drawing with straight line edges, and a compressed straight-line planar grid algorithm.

Each algorithm was applied to a given graph, resulting in eight experimental graph drawings. The experimental method was identical to that of the syntactic aesthetics experiment. Fifty-five third-year computer science students took part.

ANOVA analysis and Tukey's WSD pairwise comparison procedure were performed for both errors and response time.<sup>2</sup> The investigation gave inconclusive results, making it difficult to say that one algorithm is 'better' than another in the context of syntactic understanding of the abstract graph structure. The compressed straight-line planar grid algorithm gave slightly poorer results with respect to errors.

## 3 Semantic Preference

This experiment explored semantic preferences for graph layouts. Two Unified Modeling Language (UML) (Booch, Rumbaugh and Jacobson, 1998) diagram types were chosen as the semantic domain. Class diagrams provide a static view of a system by describing the types of objects in the system and the relationships between them. Collaboration diagrams provide a dynamic view of a system by showing the interactions between objects that send and receive messages.

Six aesthetics were evaluated in both the class and the collaboration diagrams: *minimize bends*, *minimize edge crossings*, *orthogonality*, *width of layout*, *text direction* and *font type*. Two domain-specific aesthetics were also evaluated.

This experiment used preference as the method of usability measurement. By asking whether they prefer one UML drawing to another, subjects are likely to be anticipating the use of these drawings for a software engineering task: their responses are therefore likely to be related to their perceived usefulness of the diagrams.

A UML class diagram and a UML collaboration diagram were created. The class diagram had 14 classes and 18 relationships, and the collaboration diagram had 12 objects and 17 messages. Each diagram was drawn 16 times, twice for

<sup>1</sup> Full statistical details can be found in Purchase (1997).

<sup>2</sup> Full statistical details can be found in Purchase (1998).

each aesthetic or secondary notational feature under consideration. Each pair provided a contrast for one particular layout aesthetic.

Each subject was presented with an individual evaluation booklet containing instructions, a brief tutorial on UML class diagrams, a questionnaire requesting information about the subject, eight preference questions (one for each aesthetic) and a ranking question.

Each preference question presented the participant with a contrasting pair of UML class diagrams. Subjects were asked to mark their preferred diagram and to provide an explanation for their choice. The ranking question at the end of the booklet requested that subjects rank the top three (most preferred) and the bottom three (least preferred) diagrams from the sixteen different diagrams.

Seventy student volunteers from the University of Queensland participated in the class diagram evaluation. Ninety students participated in the corresponding collaboration diagram evaluation with the same style of booklet and experimental methodology.

We concluded that the priority order of the aesthetics common to both diagram types is: arc crossing, orthogonality, information flow, arc bends, text direction, width of layout and font type.

## 4 Discussion

These three experiments have revealed much about the potential usability of graph drawings produced by automatic layout algorithms. The evidence is overwhelmingly in favour of reducing the number of arc crossings as the most important aesthetic to consider. While the results of the syntactic experiments did not highlight orthogonality as being important, in the domain of UML diagrams, this aesthetic moves up the priority list to second place: this is a clear signal that algorithms that are designed for abstract graph structures, with no consideration of their ultimate use, will not necessarily produce useful visualisations of semantic information.

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